

Evaluating Tax Reforms without Elasticities: What Bunching *Can* Identify

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Abstract

The standard sufficient statistic for evaluating income tax reforms is the elasticity of taxable income (ETI). I show that a different statistic is sufficient when a reform moves a tax bracket threshold. The bunching mass at a convex kink identifies the first-order revenue effect of the induced behavioral response, while the share of taxpayers above the kink identifies the mechanical effect. Both objects are features of the pre-reform distribution of taxable income. This result permits revenue forecasts and a test for Pareto efficiency without estimating an elasticity. An application to the first kink of the Earned Income Tax Credit illustrates how the result can incorporate heterogeneous effective tax schedules and optimization errors. The distinction is visual as well as formal: bunching reveals a vertical difference between counterfactual income distributions, while an ETI requires an unobserved horizontal difference. Moving a kink requires the former but not the latter.

1 Introduction

Progressive, piecewise linear income tax schedules are ubiquitous. A common reform changes the location of a tax bracket threshold. For example, the Tax Cuts and Jobs Act increased the income at which the top marginal rate applied to single filers from \$418,400 to \$500,000. Accurately forecasting the effect of such a reform requires predicting both the additional revenue collected mechanically and the revenue lost or gained when taxpayers change their behavior.

The standard approach to bunching methodology, first developed in Saez (2010), uses the bunching mass to generate estimates of the ETI. Recent work demonstrates that nonparametric identification of the ETI using the bunching mass and other features of the observed distribution of taxable income is not possible (Bertanha, McCallum, and Seegert 2021; Blomquist, Newey, Kumar, and Liang 2019). This limitation stems from the fact that bunching only reveals information about *vertical* differences between the CDF of taxable income under counterfactual tax schedules, but to

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estimate the ETI what we want to know is the *horizontal* difference. The various remedies proposed for this non-identification challenge can be thought of as different ways of extrapolating from the observed vertical difference to the unobserved horizontal difference.

This paper shows that for some important features of tax policy there may be no need to rely on ETI estimates in revenue forecasting. I present an approach to predicting the revenue and welfare effects of reforms that change tax brackets. The method takes advantage of a notable feature of taxpayer behavior under a progressive piecewise linear tax schedule: bunching at kinks. Such schedules induce convex kinks in taxpayer budget sets at bracket thresholds, where the marginal benefit of earning an additional dollar falls discontinuously. Consequently, for a mass of taxpayers it is optimal to earn precisely the amount of income that places them at the kink point.

The central finding is that the share of taxpayers who are bunching at a given kink identifies the revenue impact of taxpayer responses to small changes in the location of the kink. This implies that the revenue effect of moving a tax bracket threshold can be identified from the observed pre-reform distribution of taxable income alone, without relying on ETI estimates. This is a new use for empirical bunching designs that is more robust than bunching-based ETI estimation, but addresses a narrower set of policy questions.

The formal derivation is straightforward, but the intuition is somewhat subtle. Suppose a kink point is reduced by a small amount, raising the marginal tax rate on income in the window between the new and original kinks. This reform raises additional revenue from taxpayers above the kink who pay a higher rate on income within this window but whose marginal rate is unchanged. This is the mechanical effect. It also causes behavioral changes that alter tax revenue. Holding constant the size of the bunching mass, revenue falls because this mass is moved to a lower level of income at the new kink. The bunching mass will not, however, remain constant: some taxpayers are former bunchers who leave the mass, while others are new bunchers who enter it.

I show that the revenue effects associated with new and former bunchers are second order. This result obtains even though moving a kink point has a first-order effect on the size of the bunching mass. To a first approximation, taxpayers at the margin of entering or leaving the bunching mass cause no change in tax revenue. The first-order behavioral response effect is therefore identified by the pre-reform bunching mass.

I use this result in two ways. First, I discuss revenue forecasting. When a reform changes both rates and brackets, estimates of the ETI are still needed to predict the part of the behavioral response caused by rate changes. The bunching mass can be used to predict the part caused by bracket changes. Second, I use the result to derive a test for the Pareto efficiency of an observed tax schedule. In both cases, the threshold component can be evaluated using the observed distribution of taxable income.

The result concerns local movements of convex kinks. It is not a sufficient statistic for changing a marginal rate within a bracket, and discrete threshold changes generally depend on unobserved

portions of the counterfactual income distributions. Nor does the result extend without additional structure to notches, where tax liability changes discontinuously. The object identified here is the compensated behavioral response effect of a small change in the location of a convex kink.

I illustrate the practical value of these findings by revisiting Saez (2010)’s investigation of the first kink in the Earned Income Tax Credit (EITC) for single-child households between 1995 and 2004. I use this application to discuss two extensions. First, the method can accommodate settings where taxpayers face different effective tax schedules. Second, it can account for frictions in taxpayer bunching. The results show that explicitly accounting for optimization errors can substantially change the estimated behavioral response effect of moving the kink.

The paper is related to two literatures. The first characterizes optimal piecewise linear tax schedules. The sufficient-statistic result is implicit in the threshold conditions derived by Sheshinski (1989) and Apps, Van Long, and Rees (2014), but its connection to the bunching mass has gone unnoticed. The second uses bunching to estimate behavioral responses. Goff (2021) shows that bunching can identify the effect of moving a regulatory threshold on average hours. Here the outcome is government revenue under a nonlinear tax schedule. The connection to regression discontinuity designs is also useful: moving a threshold is a policy derivative with respect to the location of a discontinuity, rather than with respect to the treatment applied on either side (Dong and Lewbel 2015).

The remainder of the paper proceeds as follows. Section 2 derives the sufficient-statistic result and explains its relation to the EITC. Section 3 develops the revenue forecast, Pareto test, and fiscal externality. Section 4 presents the EITC application and the extensions needed to implement the method. Section 5 concludes.

2 The Bunching Mass as a Sufficient Statistic

2.1 Model

Consider an economy populated by taxpayer types $\theta \in \Theta$, where Θ may be multidimensional and types are continuously distributed according to F . A type θ taxpayer chooses taxable income z to solve

$$\max_z \{u(c, z; \theta) : c = z - T(z)\}. \quad (1)$$

I derive the main result under three assumptions. The first ensures that income choices exist, are unique, and change continuously under small movements of a convex kink. The second ensures that the counterfactual income distributions are smooth at the kink. The third rules out income effects in the baseline model.

Assumption 1 (Regular preferences). *For each θ , $u(c, z; \theta)$ is continuous, strictly increasing and concave in c , strictly decreasing and strictly concave in z , jointly strictly quasiconcave in (c, z) , and coercive in z .*

Assumption 2 (Regular counterfactual distributions). *For any linear tax schedule with marginal rate $t < 1$ and virtual income V , the induced CDF of taxable income $H(z; t, V)$ is continuously differentiable at the bracket threshold, with density $h(z; t, V)$.*

Assumption 3 (No income effects). *For every type θ , taxable income under a linear tax schedule is invariant to virtual income: $\partial z(t, V; \theta) / \partial V = 0$.*

Let the tax schedule have marginal rate t_0 below z^* , marginal rate $t_1 > t_0$ above z^* , and demogrant G :

$$T(z) = \begin{cases} t_0 z - G, & z \leq z^*, \\ t_1(z - z^*) + t_0 z^* - G, & z > z^*. \end{cases} \quad (2)$$

Let $z_k(\theta)$ denote the income type θ would choose under the linear tax rate t_k , and let H_k denote the corresponding CDF, for $k \in \{0, 1\}$. Under Assumptions 1–3, choices under the kinked schedule satisfy

$$z(\theta) = \begin{cases} z_0(\theta), & z_0(\theta) < z^*, \\ z_1(\theta), & z_1(\theta) > z^*, \\ z^*, & z_1(\theta) \leq z^* \leq z_0(\theta). \end{cases} \quad (3)$$

Figure 1 illustrates these choices. Some taxpayers locate in the interior of the lower bracket and choose to earn the same amount of income they would earn under the lower linear schedule. Others locate in the interior of the upper bracket and earn the same amount they would earn under the upper linear schedule.

Not all taxpayer choices can be characterized in this way. The kink induces a discontinuous drop in the marginal benefit of income at z^* . Consequently, some taxpayers' marginal-cost curves pass through this discontinuity. For each dollar of income below z^* , the marginal benefit exceeds the marginal cost, but for each dollar beyond z^* the opposite is true. These taxpayers locate at exactly z^* and are the buncers. This characterization allows for arbitrary heterogeneity in taxpayer preferences and does not require taxpayer type to be one-dimensional.

The observed CDF is assembled from the two counterfactual CDFs:

$$H(z) = \begin{cases} H_0(z), & z < z^*, \\ H_1(z), & z \geq z^*. \end{cases} \quad (4)$$

It therefore has a discontinuity at z^* . The size of this discontinuity is the bunching mass,

$$B \equiv \Pr\{z_1(\theta) < z^* < z_0(\theta)\} = H_1(z^*) - H_0(z^*). \quad (5)$$

Figure 2 shows both the construction of the observed CDF and the revenue associated with the tax schedule. Importantly, B and $1 - H_1(z^*)$ are features of the observed distribution even though the complete functions H_0 and H_1 are not observed.

The CDF representation is useful because tax revenue can be read as an area. Income to the left of the threshold is taxed at t_0 , while income above the threshold is additionally taxed at $t_1 - t_0$. In Figure 2, the relevant areas lie above the observed CDF. The figure also provides some intuition for why bunching occurs: taxpayers who would make distinct choices under linear schedules lump together at z^* when facing the kinked schedule.

2.2 A graphical derivation

Without loss of generality, set $G = 0$. Revenue under the tax schedule is

$$\begin{aligned}
 R(z^*) = & \underbrace{t_0 \int_0^{z^*} z h_0(z) dz}_{\text{revenue below the kink}} + \underbrace{t_0 z^* [H_1(z^*) - H_0(z^*)]}_{\text{revenue from bunchers}} \\
 & + \underbrace{\int_{z^*}^{\infty} [t_1(z - z^*) + t_0 z^*] h_1(z) dz}_{\text{revenue above the kink}}.
 \end{aligned} \tag{6}$$

Differentiating equation (6) gives the main result directly. A graphical derivation provides more intuition.

Consider a reform that lowers the threshold from z^* to z' . Taxpayers fall into five groups. Persistent lower- and upper-bracket taxpayers do not change their behavior. Persistent bunchers move from z^* to z' . Former bunchers leave the bunching mass and locate between the new and old thresholds. New bunchers previously located between the two thresholds and join the mass at z' . Figure 3 shows the reform, the five groups, and the associated revenue effects.

The mechanical effect is the additional revenue from taxing income in $(z', z^*]$ at the higher rate. The behavioral response effect is the revenue lost when taxpayers reduce their income in response. For a finite reform, both effects depend on the unobserved portions of the counterfactual CDFs between z' and z^* . In the limiting case, this dependence disappears. The mechanical region becomes a rectangle with height $1 - H_1(z^*)$; the behavioral region becomes a rectangle with height $H_1(z^*) - H_0(z^*)$. The revenue effects of new and former bunchers are second order because taxpayers at the margin of entering or leaving the bunching mass generate no first-order change in revenue.

More formally, the mechanical effect of a finite decrease from z^* to z' is

$$(t_1 - t_0) \left\{ (z^* - z') [1 - H_1(z^*)] + \int_{z'}^{z^*} (z - z') h_1(z) dz \right\}. \tag{7}$$

The first term is paid by taxpayers who were already above the old kink. The integral is paid by former bunchers who locate between the new and old thresholds after the reform. The behavioral

response effect is

$$-t_0 \left\{ \int_{z'}^{z^*} (z - z') h_0(z) dz + (z^* - z') [H_1(z') - H_0(z^*)] + \int_{z'}^{z^*} (z^* - z) h_1(z) dz \right\}. \quad (8)$$

The three terms correspond to new bunchers, persistent bunchers, and former bunchers. Equations (7) and (8) are not identified for a finite reform because H_0 and H_1 are not observed throughout (z', z^*) . They are included to make clear what disappears in the limit.

Let $z^* - z'$ approach dz^* . The integral terms in equations (7) and (8) become second order. The persistent-buncher term becomes $-t_0 B dz^*$, while the revenue collected from taxpayers above the kink becomes $(t_1 - t_0)[1 - H_1(z^*)] dz^*$. Both heights are observed at the original threshold. This yields Theorem 1.

Theorem 1 (Sufficiency of the bunching mass). *Under Assumptions 1–3, the first-order revenue effect of decreasing the kink point is*

$$-R'(z^*) = \underbrace{(t_1 - t_0) [1 - H_1(z^*)]}_{\text{mechanical effect}} - \underbrace{t_0 \overbrace{[H_1(z^*) - H_0(z^*)]}^{\text{bunching mass}}}_{\text{behavioral response effect}}. \quad (9)$$

The bunching mass is a sufficient statistic for the revenue effect of the behavioral response. The probability of locating above the kink is a sufficient statistic for the mechanical effect.

Importantly, Theorem 1 applies to convex kinks, at which tax liability is continuous. It does not extend without additional structure to a notch. At a notch, taxpayers at the margin may switch between the threshold and a discrete upper income, and the resulting change in tax payments depends on that destination income. The pre-reform bunching mass does not identify this information.

2.3 Relation to the elasticity of taxable income

The bunching mass in equation (9) is the same object that the standard bunching method uses to estimate the ETI. That method divides the bunching mass by a normalizing term,

$$\mathcal{H} \equiv z^* \int_{t_0}^{t_1} \frac{h(z^*; t)}{1 - t} dt, \quad (10)$$

to recover a weighted average of elasticities. The observed income distribution does not identify $\mathcal{H}$, and so it does not identify the elasticity (Bertanha, McCallum, and Seegert 2021; Blomquist, Newey, Kumar, and Liang 2019).

Under a single-crossing condition, the relationship can be written as

$$H_1(z^*) - H_0(z^*) = \int_{t_0}^{t_1} \bar{\varepsilon}^c(z^*; t) \frac{z^* h(z^*; t)}{1 - t} dt, \quad (11)$$

where $\bar{\varepsilon}^c(z^*; t)$ is the average compensated ETI among taxpayers who would locate at z^* under the counterfactual linear rate t . Equation (11) shows that the bunching mass is a function of both taxpayer elasticities and unobserved densities of taxable income. Thus, the bunching mass alone cannot identify the ETI.

In order to obtain a weighted average of elasticities, it is necessary to have an estimate of the normalizing term $\mathcal{H}$. Recent work has called attention to the fact that the observed distribution of taxable income cannot nonparametrically identify this term (Bertanha, McCallum, and Seegert 2021; Blomquist, Newey, Kumar, and Liang 2019). The proposed identification strategies impose assumptions about how the density of taxable income at z^* varies across counterfactual linear tax schedules.

Figure 4 depicts this problem in log income. The red arrow in both panels is the vertical difference revealed by bunching. The horizontal arrow is inferred by imposing a mapping between that vertical difference and taxpayer responses. A different estimated elasticity changes the horizontal response while leaving the observed bunching mass unchanged. Moving a kink requires the vertical difference. Estimating an ETI requires the unobserved horizontal difference.

3 Evaluation without Elasticities

3.1 Revenue forecasting

Let (t_0, t_1, z^*) be the current tax parameters and consider a reform to $(t_0 + \Delta t_0, t_1 + \Delta t_1, z^* - \Delta z^*)$, where $\Delta z^* > 0$. A first-order revenue forecast is

$$\begin{aligned} \Delta R \approx & \underbrace{(t_1 - t_0) [1 - H_1(z^*)] \Delta z^*}_{\text{mechanical effect of threshold change}} - \underbrace{t_0 [H_1(z^*) - H_0(z^*)] \Delta z^*}_{\text{behavioral effect of threshold change}} \\ & + \underbrace{\frac{\partial R}{\partial t_0} \Delta t_0 + \frac{\partial R}{\partial t_1} \Delta t_1}_{\text{rate-change effects}}. \end{aligned} \quad (12)$$

Equation (12) makes clear that not every aspect of tax reform evaluation requires knowledge of taxpayer elasticities. The part of the reform caused by tax bracket changes can be estimated relying only on features of the pre-reform distribution of taxable income. On the other hand, predicting the revenue effect of changing rates within a bracket does require information about the ETI.

To understand the value of this result, it is helpful to compare equation (12) to the common practice of approximating a behavioral response by multiplying the change in a taxpayer's effective marginal rate by taxable income and an assumed ETI. This elasticity-based approach treats bracket changes as another way of changing marginal rates over some range of income and attempts to use elasticities to approximate the response to those rate changes.

My results show that this reliance on elasticities is unnecessary for the threshold component of the reform. Moreover, the elasticity-based approximation does not generally coincide with equation (12), even if the correct elasticities are used. Estimates of the ETI remain necessary for the rate-change terms. The point is that a reform which changes both rates and brackets contains separate components, and the bunching mass directly identifies the component caused by the bracket change.

3.2 A test for Pareto efficiency

Theorem 1 can also be used to derive a simple test for the Pareto efficiency of an observed tax schedule. Consider a reform that increases the bracket threshold z^* . This reform expands the budget set and so, by revealed preference, all taxpayers weakly prefer the post-reform tax schedule. If it also causes a net increase in tax revenue, it constitutes a Pareto improvement.

Theorem 2 (Test for Pareto efficiency). *Under Assumptions 1–3, Pareto efficiency of the two-bracket tax schedule requires*

$$R'(z^*) \leq 0. \quad (13)$$

If $R'(z^) = 0$, it also requires*

$$R''(z^*) = t_1 h_1(z^*) - t_0 h_0(z^*) \leq 0. \quad (14)$$

Both conditions are testable using the pre-reform distribution of taxable income.

If $R'(z^*) > 0$, or if $R'(z^*) = 0$ and $R''(z^*) > 0$, there exists some sufficiently small increase in the kink point that increases tax revenue. As noted above, this is a Pareto-improving reform. When $R''(z^*) = 0$, the second-order test is inconclusive. Theorem 1 shows that the first derivative is identified by the observed distribution of taxable income. The density terms in the second derivative are also identified because they are limit points of the observed income density from below and above the kink.

This test is an analogue of the test for local Laffer effects proposed by Werning (2007). Prior implementations require a model of taxpayer behavior and calibrated parameters. By contrast, the test here accounts for the behavioral response to changing a bracket threshold without information beyond the observed distribution of taxable income. This robustness comes at the cost of scope, since the test is uninformative about other tax reforms of interest.

3.3 The fiscal externality

The fiscal externality is the behavioral response effect divided by the mechanical effect. It measures the revenue lost through behavior per dollar raised mechanically.

Corollary 1 (The fiscal externality is identified). *Under Assumptions 1–3, the fiscal externality*

from marginally decreasing the kink is

$$FE(z^*) = -\frac{t_0 [H_1(z^*) - H_0(z^*)]}{(t_1 - t_0) [1 - H_1(z^*)]}, \quad (15)$$

and is identified by the pre-reform distribution of taxable income.

The fiscal externality provides an intuitive summary of the revenue effects of the reform: it is the revenue loss due to the behavioral response per dollar raised through the mechanical effect. Equation (15) can also be used to characterize Rawlsian and Bergsonian welfare conditions, although those conditions require additional assumptions about social welfare weights and the private welfare effects experienced by bunchers. In the empirical application below, I use the fiscal externality simply as a summary of the revenue effects.

4 Empirical Application

This section applies the ideas discussed above to revisit an empirical application from Saez (2010). Saez uses the IRS Individual Public Use Tax Files to investigate bunching at the first kink point of the EITC between 1995 and 2004. I use the same sample to illustrate how the bunching mass can be used for policy analysis.

During this period, the EITC provided a 34 percent subsidy on the marginal dollar of family earnings in a single-child household below the first kink. The marginal subsidy fell to zero above this threshold, introducing a convex kink in the household's budget constraint. The statutory federal schedule is not, however, the complete tax schedule faced by these households. Federal and state income taxes, payroll taxes, state EITC supplements, and welfare benefit phaseouts all affect the relevant marginal rates.

4.1 Recovering the bunching mass

Before applying the results above, it is necessary to address frictions in taxpayer choices. Real-world taxable income distributions do not support the assumption of sharp bunching. As Saez (2010) noted, some groups of EITC recipients exhibit a diffuse lump in the density around the first kink rather than a mass point precisely at it. I assume that observed income z is the product of optimal income y and an optimization error ξ :

$$z = y\xi, \quad \tilde{z} = \tilde{y} + \tilde{\xi}, \quad (16)$$

where a tilde denotes logs. Following Cattaneo, Jansson, Ma, and Slemrod (2018), I assume that $\tilde{\xi}$ is mean-zero normal with variance σ^2 . If the latent distribution of optimal log income features a non-zero bunching mass at the kink, then both the latent distribution and σ^2 are semiparametrically

identified by the observed distribution.

I approximate the two counterfactual latent CDFs using eighth-order polynomials and estimate their coefficients by constrained minimization of simulated least squares. I apply this filtering method independently to unmarried employees, married employees, unmarried self-employed taxpayers, and married self-employed taxpayers. The estimated bunching mass is less than one percent in three of the four groups, but for unmarried self-employed taxpayers it is 7.93 percent. Figure 5 shows the estimated latent CDF for this group alongside the empirical CDF of log income and the simulated CDF implied by the fitted model. The simulated CDF closely tracks the empirical CDF.

The estimated shares of taxpayers above the kink are 85.43 percent for unmarried employees, 74.76 percent for unmarried self-employed taxpayers, 97.32 percent for married employees, and 95.13 percent for married self-employed taxpayers. I use these estimates, together with the estimated bunching masses, as inputs for the policy analysis below.

4.2 Heterogeneous effective tax schedules

An ideal analysis of the fiscal effects of moving the EITC kink would account for all the ways that taxable income choices influence government revenue and expenditure. Thus, the statutory marginal tax rates from the EITC schedule are not the relevant rates for this analysis. Suppose that a type θ taxpayer faces rates $t_0(\theta)$ and $t_1(\theta)$ on either side of the common threshold. The first-order revenue effect is

$$\begin{aligned}
 -R'(z^*) = & \underbrace{\mathbb{E}[t_1(\theta) - t_0(\theta) \mid z(\theta) > z^*] [1 - H_1(z^*)]}_{\text{mechanical effect}} \\
 & - \underbrace{\mathbb{E}[t_0(\theta) \mid z_1(\theta) < z^* < z_0(\theta)] [H_1(z^*) - H_0(z^*)]}_{\text{behavioral response effect}}.
 \end{aligned} \tag{17}$$

I use TAXSIM to determine effective marginal tax rates and incorporate state EITC supplements, payroll taxes, and approximations to AFDC/TANF and SNAP phaseouts. Contrary to what statutory federal rates would imply, effective marginal rates on either side of the EITC kink are positive for the average taxpayer in most groups. The negative marginal rate created by the EITC is offset by other taxes and welfare benefit phaseouts. Only for unmarried self-employed taxpayers is the average rate just below the kink negative.

For unmarried employees, the estimated local average rates are 0.008 below the kink and 0.314 above it. For unmarried self-employed taxpayers they are -0.007 and 0.283. The corresponding estimates are 0.164 and 0.342 for married employees and 0.059 and 0.276 for married self-employed taxpayers. The estimated size of the kink is therefore smaller than the statutory EITC rates imply for every group.

For a \$100 reduction in the first kink, the estimated behavioral response effect is small relative to the mechanical effect. For all EITC-eligible taxpayers, the mechanical effect is approximately

\$38.39 million per year and the behavioral response effect is approximately -\$33,600, yielding a fiscal externality of -0.09 percent. These calculations require the additional assumption that the average rates faced by taxpayers near the kink are also the relevant rates for the bunchers and taxpayers entering the two terms in equation (17).

For the three reported subgroups, the estimated fiscal externalities are 0.04 percent for unmarried taxpayers, -0.03 percent for self-employed taxpayers, and 0.25 percent for taxpayers who are both unmarried and self-employed. The behavioral response effect is positive for the groups with a negative average marginal rate below the kink. Reducing the kink moves bunchers to a lower level of taxable income and thereby reduces the subsidy they receive.

4.3 Optimization errors and discrete reforms

The preceding calculations pertain to the hypothetical revenue obtained by taxing unobserved optimal income choices. Actual taxes are levied on observed taxable income, which includes the effect of optimization error. When optimization error is present, bunching taxpayers are scattered around the kink, with some locating in the lower bracket and others in the upper bracket. Obtaining the behavioral response effect therefore requires weighting the responses of bunchers above and below the kink by their corresponding marginal tax rates. The first-order revenue effect is

$$\begin{aligned}
 -R'(z^*) = & (t_1 - t_0) [1 - H_z(z^*)] \\
 & - \{t_0 F_\xi(1) \mathbb{E}[\xi \mid \xi \leq 1] + t_1 [1 - F_\xi(1)] \mathbb{E}[\xi \mid \xi > 1]\} [H_y(z^*; t_1) - H_y(z^*; t_0)].
 \end{aligned} \tag{18}$$

There are two important differences from the calculation based on optimal income choices. First, the mechanical effect uses the observed share of taxpayers above the kink. Second, the behavioral response effect uses an effective marginal tax rate that reflects the variation in the observed location of bunchers. In the EITC application, accounting for optimization errors raises the fiscal externality for all EITC-eligible taxpayers from 0.09 percent to 0.55 percent in absolute value. For unmarried self-employed taxpayers, the estimate changes from a positive 0.25 percent to a negative 4.67 percent. The corresponding estimates are -1.13 percent for unmarried taxpayers and -1.03 percent for self-employed taxpayers.

These differences primarily result from the effective marginal tax rate of bunchers. Some bunchers are observed below the kink and others above it, so their behavioral responses must be weighted by different marginal rates. This matters especially in the EITC application because the estimated average rate immediately below the kink is small for most groups.

Theorem 1 is a local result. Identification of the revenue effect of a discrete decrease in the kink generally requires knowledge of unobserved values of H_1 between the old and new kink points. I estimate these values using the polynomial approximation obtained from the filtering procedure. The resulting estimates are close to the first-order calculations, but this extrapolation exercise is not justified by the semiparametric identification argument. Higher-order approximations and

partial-identification methods are discussed in the online appendix.

Finally, the baseline calculations exclude income and labor force participation effects. The bunching mass remains sufficient for the component of the fiscal externality attributable to substitution effects, but it does not identify the income or participation components. Accounting for these effects requires external evidence about their likely magnitude.

5 Conclusion

The concept of the elasticity of taxable income as a sufficient statistic for the revenue and welfare effects of taxation is powerful. However, it is not always necessary to have an estimate of the ETI in order to predict the impact of a reform. Under basic theoretical restrictions, the bunching mass at a tax bracket threshold is itself a sufficient statistic for the compensated behavioral response effect of a local movement of the threshold. This use of the bunching mass is fully nonparametric and requires no information about taxpayer preferences beyond the observed distribution of taxable income generated by their choices.

Building on this result, I develop an approach to revenue forecasting and an empirical test for Pareto efficiency. The EITC application motivates extensions that incorporate important features of real-world policy settings. In particular, the results show that correctly accounting for optimization errors can have a meaningful impact on the estimated revenue effects of moving a bracket threshold.

Reforms that change bracket thresholds are not the only reforms of interest. The method proposed here is therefore best viewed as a complement to, rather than a substitute for, elasticity estimation. The bunching mass reveals a vertical difference between counterfactual income distributions that is sufficient for analyzing a movement of the kink. Analysis of rate changes within brackets requires an unobserved horizontal difference and therefore continues to require an elasticity or additional structure.

Saez (2010) presented a tantalizing idea: bunching in the observed distribution of taxable income can teach us about the response to taxation even absent variation in the tax schedule over time or across groups. Recent work has documented substantial challenges to identification and interpretation of bunching-based ETI estimates. By shifting the focus away from the ETI, the bunching mass can instead be used to identify a different policy-relevant parameter with a clear interpretation. The focus of the prior literature on the ETI is understandable given its importance in public economics. However, that focus has come at the cost of overlooking another valuable application of the bunching mass.

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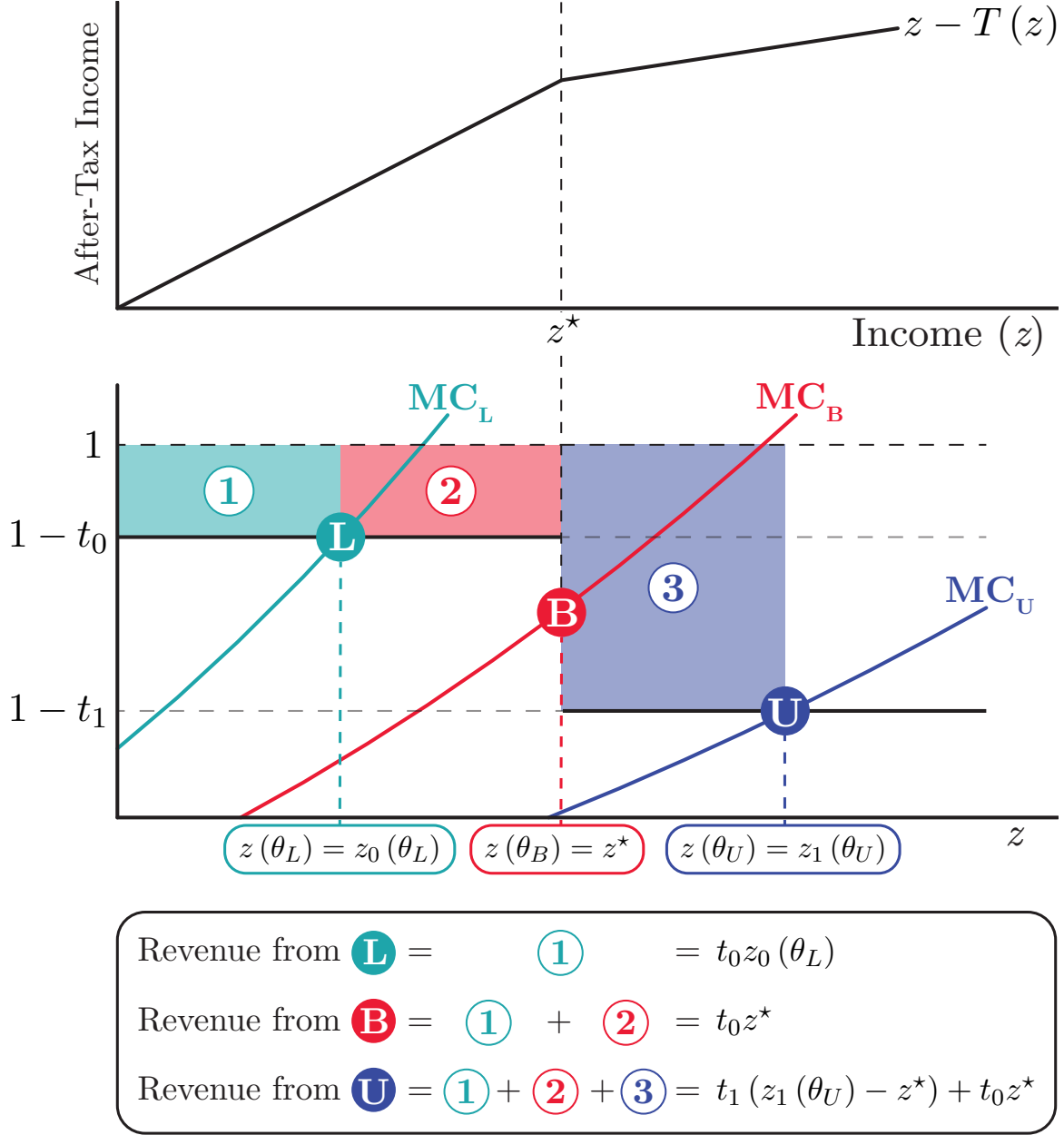


Figure 1: Taxpayer choices and tax payments. *Notes:* The upper panel depicts a two-bracket progressive tax schedule. The lower panel shows the marginal benefit of income and three representative marginal-cost curves. Lower- and upper-bracket taxpayers choose an interior income. A buncher's marginal-cost curve passes through the discontinuity in marginal benefits, so the taxpayer locates at z^* . The tax payment generated by each choice is the area between one and the marginal-benefit schedule up to that choice.

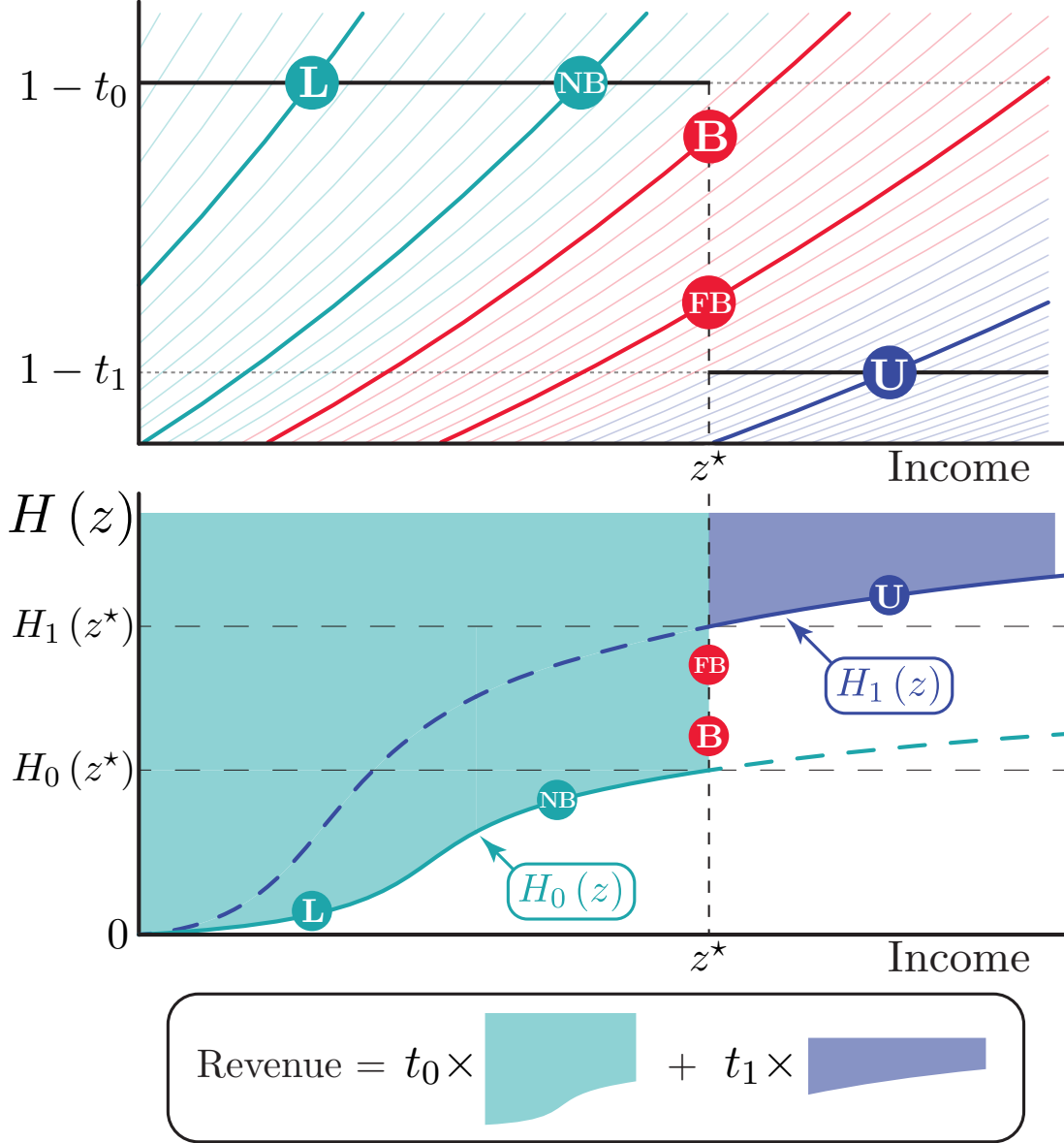


Figure 2: The pre-reform distribution of taxable income and tax revenue. *Notes:* Below z^* , the observed CDF coincides with H_0 ; above z^* , it coincides with H_1 . The vertical discontinuity at the kink is the bunching mass. The area above the observed CDF and to the left of z^* is income taxed at t_0 ; the area to the right is income taxed at t_1 .

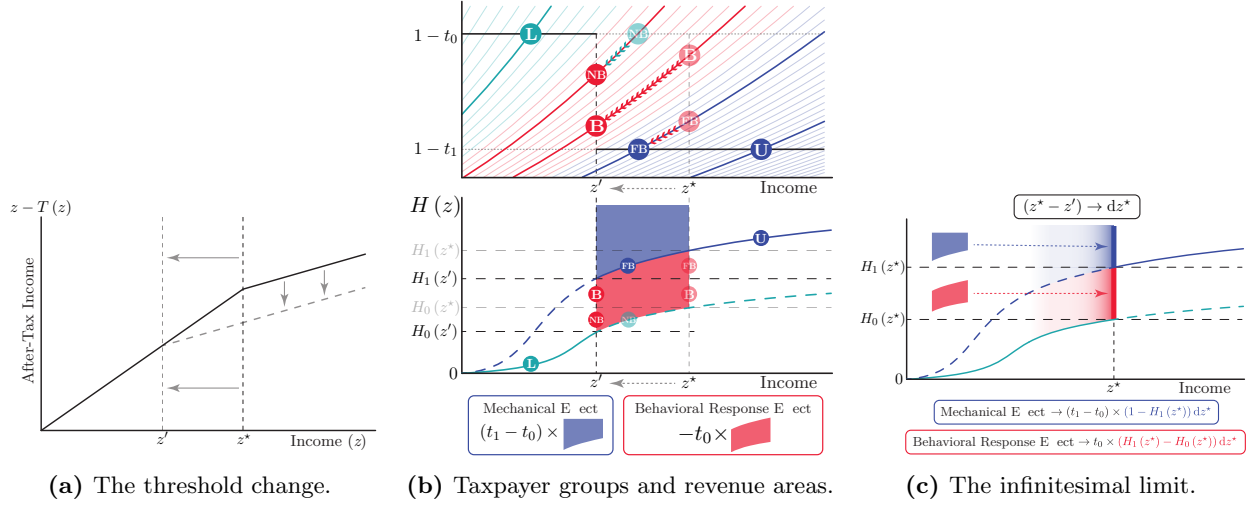


Figure 3: Revenue effects of moving the kink. *Notes:* Lowering the threshold raises the marginal tax rate over $(z', z^*]$. The blue region is the mechanical effect. The red region is the behavioral response effect. For a finite reform, the two effects depend on counterfactual portions of H_0 and H_1 that are not observed. As $z^* - z'$ approaches zero, the regions collapse to rectangles whose heights are observed at the original kink.

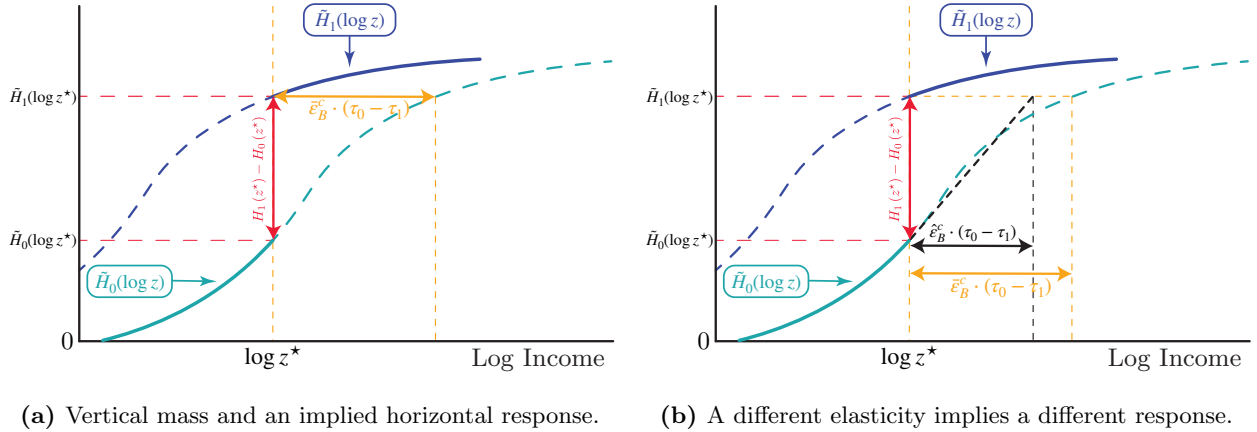


Figure 4: What bunching does and does not identify. *Notes:* The red arrow is the observed vertical difference $\tilde{H}_1(\log z^*) - \tilde{H}_0(\log z^*)$. The horizontal arrows are responses inferred by imposing an elasticity-based mapping. Alternative mappings can rationalize the same vertical difference. A movement of the kink requires the vertical difference; an ETI requires the horizontal difference.

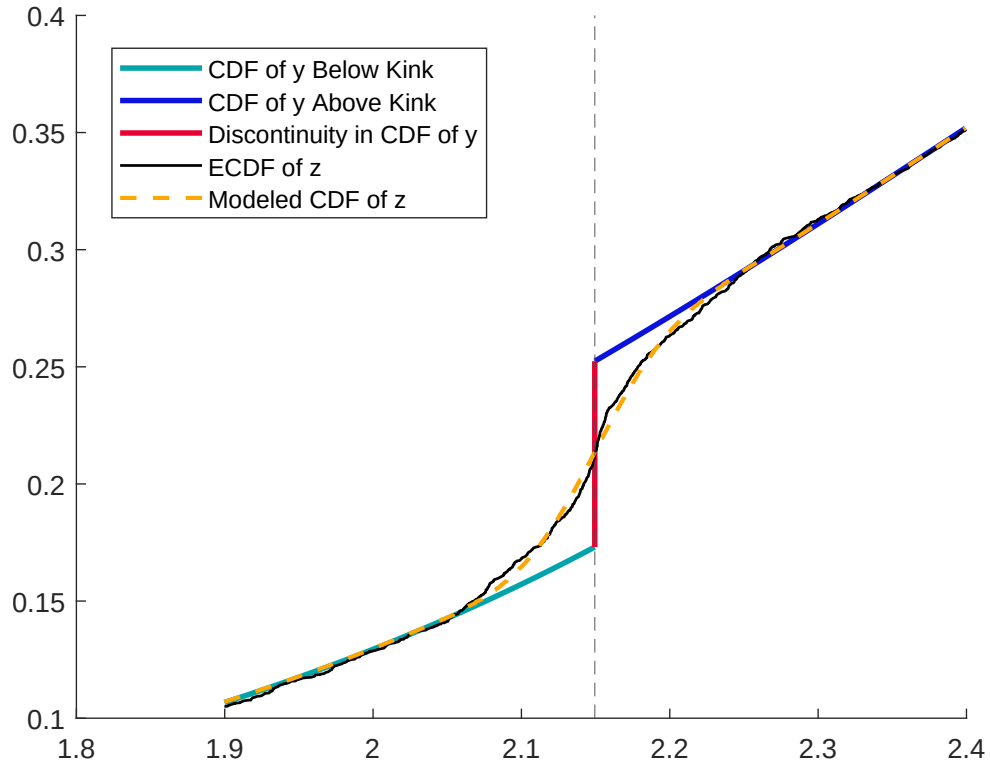


Figure 5: Estimated CDF of optimal income choices. *Notes:* The figure reports the empirical CDF of log income for unmarried self-employed taxpayers, the recovered latent CDF, and the simulated CDF implied by the fitted model. The estimated bunching mass is 7.93 percent and the estimated share above the kink is 74.76 percent. Using heterogeneous effective tax rates, the frictionless fiscal externality for this group is 0.25 percent. Accounting for optimization errors changes its sign and raises its magnitude to 4.67 percent in absolute value.